

1 Summary

I chose the **Philippines** for having a moderately higher GDP than other developing nations. It is also a newly industrialized country (NIC) having a mostly agrarian economy.¹ To study where industrial zones are located, I assembled a spatial dataset of 102 zones from government and international sources, linked it to satellite-derived and infrastructure data at the municipality level, and compared zone versus non-zone locations descriptively. The analysis suggests that industrial zones are systematically located in municipalities with higher population density, greater nighttime economic activity, and closer proximity to primary road networks. This is a partial solution - further work is needed to verify data accuracy and address missing datasets.

2 Background and Motivation

The Philippines enacted the Special Economic Zone Act of 1995 (Republic Act 7916), establishing the Philippine Economic Zone Authority (PEZA) as the coordinating body for industrial zone development.⁵ PEZA currently administers over 300 economic zones of various types, with manufacturing and agro-industrial zones being the primary vehicles for industrial development and foreign direct investment attraction (Ortega et al., 2015).

Understanding *where* these zones are placed - and how those locations differ from non-zone areas - is foundational to studying whether place-based industrial policy shapes local economic outcomes. This analysis constructs the spatial dataset necessary for such an inquiry.

3 Prior Work

Ortega, Aciego, and Hermida (2015) conducted the most directly relevant prior study, mapping emergent mega-regions in the Philippines using municipal-level census data (1990, 2000, 2010) overlaid with PEZA zone locations and road networks. They identified 11 mega-regions and demonstrated that PEZA zones cluster in high-density municipalities along major transport corridors. However, their work did not produce a publicly available geocoded dataset of zone locations. The present analysis fills this gap by constructing a reproducible spatial dataset with coordinates for 102 industrial zones.

4 Sources Reviewed

Twenty data sources were identified and documented in `sources.csv`. Key sources: (1) PEZA Manufacturing Zones - 74 zones with names, locations, area; (2) PEZA Agro-Industrial Zones - 22 operating + 6 proclaimed; (3) World Bank CIIP SEZ Database - 47 zones with GPS coordinates and establishment years; (4) GADM Philippines v4.1 - municipality boundaries; (5) VIIRS Nighttime Lights 2018 - economic activity proxy; (6) WorldPop Population 2020 - population density; (7) GHSL Built-up Surface 2020 - urbanization; (8) OpenStreetMap Philippines - roads, waterways, transport; (9) OurAirports - airport locations. All sources accessed May 2026.

5 Results

- **GitHub Repository:** <https://github.com/AGR19/SEZ-Spatial-Data-Research>
- **Live Interactive Dashboard (Bonus):** <https://agr19.shinyapps.io/philippines-industrial-zones/>
- **R-Generated Figures:** <https://github.com/AGR19/SEZ-Spatial-Data-Research/tree/main/figures>
- **ArcGIS Pro Maps:** <https://github.com/AGR19/SEZ-Spatial-Data-Research/tree/main/ResultMapsArcGIS>

Note: The Shiny dashboard is a bonus deliverable beyond assignment requirements, demonstrating interactive spatial data exploration.

6 Methods

Spatial unit: Municipality (GADM level 2, 1,628 units). **CRS:** EPSG:4326 (WGS 84) for storage; EPSG:3123 (PRS 92) for distance calculations. **Tools:** R 4.5.1 + ArcGIS Pro (with Living Atlas).

Step	What was done	So what
Zone construction	Fuzzy-matched names across 5 sources, geocoded 27 unmatched to centroids	First geocoded dataset of 102 Philippine industrial zones
Spatial merge	Extracted raster stats per municipality; computed distances to airports/roads	Municipality-level dataset linking zone presence to characteristics
Comparison	Welch's t-tests, correlations, logistic regression	Quantified zone vs non-zone differences on 5 variables
Dynamic extension	Compiled est. years for 73 zones; identified VIIRS/WorldPop as panel data	Documented feasibility of staggered DiD design

7 Findings

The strongest predictor of zone presence is proximity to primary roads (logistic regression coefficient = -0.92). This is consistent with zones being placed to leverage existing infrastructure.

8 Conclusions

The data suggest that Philippine industrial zones reinforce existing urban-industrial corridors rather than creating new development poles in remote areas. This is descriptive, not causal - zones were intentionally placed near infrastructure.

Characteristic	Zone (N=56)	Non-Zone (N=1,572)	Difference	p-value
Population density (persons/km ²)	2,570	646	+1,924	0.002***
Nighttime lights (nW/cm ² /sr)	3.52	0.55	+2.97	<0.001***
Built-up fraction	0.054	0.013	+0.041	<0.001***
Distance to primary road (km)	3.0	8.5	-5.5	<0.001***
Distance to airport (km)	34.8	42.6	-7.8	0.080*

Table 1: Zone vs Non-Zone Municipality Comparison (Welch’s t-test). Significance: *** p<0.01, * p<0.10

A causal analysis using staggered difference-in-differences, exploiting variation in establishment timing (1974–2014), would be the natural next step.

This is a partial solution. Next steps include verifying data accuracy, handling missing datasets, and obtaining complete PEZA zone coordinates via Freedom of Information request.

9 Limitations and Future Directions

Limitations:

1. Selection bias - zones were placed deliberately near infrastructure, so observed differences reflect government site selection, not zone effects.
2. Geocoding imprecision - 27 zones lack GPS coordinates and were approximated to municipality centroids (several km error possible).
3. Temporal gaps - zone data is from 2017, nightlights from 2018, population from 2020. I assume municipality characteristics are stable over this 2-3 year window.
4. Island geography - I used Euclidean (straight-line) distances, which do not account for water crossings between islands in the Philippine archipelago.
5. Missing zone types - IT Parks (262) and Tourism Zones (19) are not individually listed in any publicly available dataset I could find.

What I would do next with more time:

- File a Freedom of Information (FOI) request directly to PEZA to obtain the complete zone database with GPS coordinates and polygon boundaries.
- Download VIIRS nighttime lights for every year from 2012 to 2023, then run a proper difference-in-differences panel regression using the 73 zones with known establishment dates.
- Try again to access PSA OpenStat for municipality-level employment and firm count data - perhaps through a different browser or by contacting PSA directly.
- Extract land values from Bureau of Internal Revenue (BIR) zonal value gazette PDFs, which are published per revenue district but not compiled digitally.
- Build a matched control group using propensity scores on pre-treatment population density, distance to Manila, and baseline nightlights.

10 Pain Points

1. Some government websites (even the one cited on Wikipedia itself) were not operational.^{1,2}
2. Philippines Statistics Authority’s OpenStat³ websites - some were not working, only a few worked. I could do queries on the table but I was not able to download the csv files. (Showed Cloudflare Error 520 or PXWeb Errors when I open some datasets. Wish I had this data. The quality and the micro/macroeconomic segments of datasets were too good!)
3. GADM⁴ download links not working. Had to download it via R geodata package.

11 Software and Versions

R 4.5.1; sf 1.1.0; terra 1.9.11; exactextractr 0.10.1; dplyr 1.1.4; ggplot2 3.5.2; stringdist 0.9.17; geodata 0.6.9; corrplot 0.95; shiny 1.13.0; leaflet 2.2.3; plotly 4.12.0; ArcGIS Pro 3.x.

12 AI Disclosure

I used Claude Opus 4.6 with high effort (1M tokens) on Claude Code.

13 Technical Appendix

Full code, data documentation, and reproducibility instructions available at:
<https://github.com/AGR19/SEZ-Spatial-Data-Research>

References

- Ortega, A.A., Aciego, J.M.A.E., & Hermida, M.C.H. (2015). Mega-regions in the Philippines: Accounting for special economic zones and global-local dynamics. *Cities*, 48, 130–139. <https://doi.org/10.1016/j.cities.2015.07.002>
- Philippine Economic Zone Authority (2017). *List of Operating Manufacturing Economic Zones*. Retrieved from <https://www.peza.gov.ph/sites/default/files/1.xls>

¹ Wikipedia contributors. *Economy of the Philippines*. https://en.wikipedia.org/wiki/Economy_of_the_Philippines

² <https://www.peza.gov.ph/index.php/economic-zones/list-of-economic-zones/operating-economic-zones>

³ <https://openstat.psa.gov.ph/>

⁴ https://gadm.org/download_country.html

⁵ Republic of the Philippines (1995). *Republic Act No. 7916: Special Economic Zone Act of 1995*.